

Distribution of academic research funds: a case of Japanese national research grant

Sotaro Shibayama

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Abstract Drawing on a database of the competitive research funds in the Japanese academia, this study examines the distribution of research grants at the university and individual levels. The data indicates high inequality at the university level and slightly lower inequality at the individual level. Over the last three decades, the total grant budget has greatly increased and an increasing number of researchers have received the funds. Simultaneously, large-size grants have become more common and multiple awarding (i.e., one researcher receives more than one grant simultaneously) has become more frequent. These changes taken together, the level of inequality has not been changed substantially. The extent of inequality largely differs between scientific fields; especially high in basic natural sciences and relatively low in social sciences. A close examination of inequality over researchers' career indicates different patterns of transition between fields and cohorts. Finally, both at the university and individual levels, the funding distribution is found more unequal than the distribution of publications as an output indicator.

Keywords Research grant · Funding · Inequality · Academia · University

JEL Classification H81 · I23 · I28 · D63

S. Shibayama (✉)

Department of Economics, The University of Turin, Via Po 53, 10124 Turin, Italy
e-mail: shibayama@00.alumni.u-tokyo.ac.jp

S. Shibayama

School of Public Policy, Georgia Institute of Technology, 685 Cherry Street,
Atlanta, GA 30332-0345, USA

S. Shibayama

Research Center for Advanced Science and Technology, The University of Tokyo,
4-6-1 Komaba, Meguro-ku, Tokyo 153-8904, Japan

Introduction

Academic research is essentially underpinned by funding from public sources. Research activities in academic institutions, regarded as the foundation of innovation system (e.g., Etzkowitz and Leydesdorff 2000), have been supported by public funds, but increasing pressure for accountability and tightening budget has urged greater efficiency of funding (Geuna and Martin 2003; OECD 1997). From the perspective of individual researchers and academic institutions, stable access to research funds is vital amid the intensifying competition. Consequently, previous literature has examined various aspects of the funding system, such as peer-review (Kotchen et al. 2004), suitable grant size (Baumeister and Bacharach 1997), and support for junior researchers (Wadman 1997). Among others, the equality of funding distribution has been of great interest. Many countries distribute research funds more or less on the basis of the performance of researchers or academic institutions (Geuna and Martin 2003). Considering the substantial disparity in the performance of researchers (Lotka 1926), the performance-based funding inevitably results in highly unequal distribution. This funding strategy seems reasonable in that it can provide a direct incentive to improve the output of research. However, such a system could decrease the diversity of research subjects and discourage challenging topics (Geuna and Martin 2003), compromising scientific progress in the long term. Excessive concentration of funds in the hands of specific scientists may also be inefficient due to diminishing returns. To advance this line of argument, funding data has to be closely examined. However, funding data is often difficult to access in a comprehensive manner, so previous literature on research funding has tended to draw on fairly limited data or focus on qualitative aspects. Importantly, few studies have examined the funding distribution at the level of individual researchers, while some literature has studied university-level distribution (e.g., Hicks and Katz 2009). This is a critical pitfall since individual- (or project-) level funding constitutes a significant part of the competitive funding system in many countries. To fill in this gap, this study aims to provide quantitative information on funding distribution, especially focusing on the individual level. To this end, I draw on a national grant database in the Japanese academia, which covers the majority of the research grants awarded during the last four decades.

The remainder of this article is structured as follows. “[Data and methods](#)” section explains the data and methods of this study with the overview of the funding system in Japanese academia. “[University level distribution](#)” section briefly describes the university-level funding distribution. “[Individual level distribution](#)” section analyzes the individual-level distribution from various perspectives. “[Inequality in output](#)” section compares the distributions of funding and publications. Finally, “[Discussion](#)” section summarizes the results and discusses the direction of future research.

Data and methods

Overview of the funding system in Japanese universities

Japanese universities consist of 86 national universities, 95 other public universities (operated by cities or prefectures), and 597 private universities in 2010 (Fig. 1a illustrates the transition of the number of universities). Among these, the national universities are the main player of academic research. While several private and public universities are also considered research-intensive, many private universities are education-oriented. For

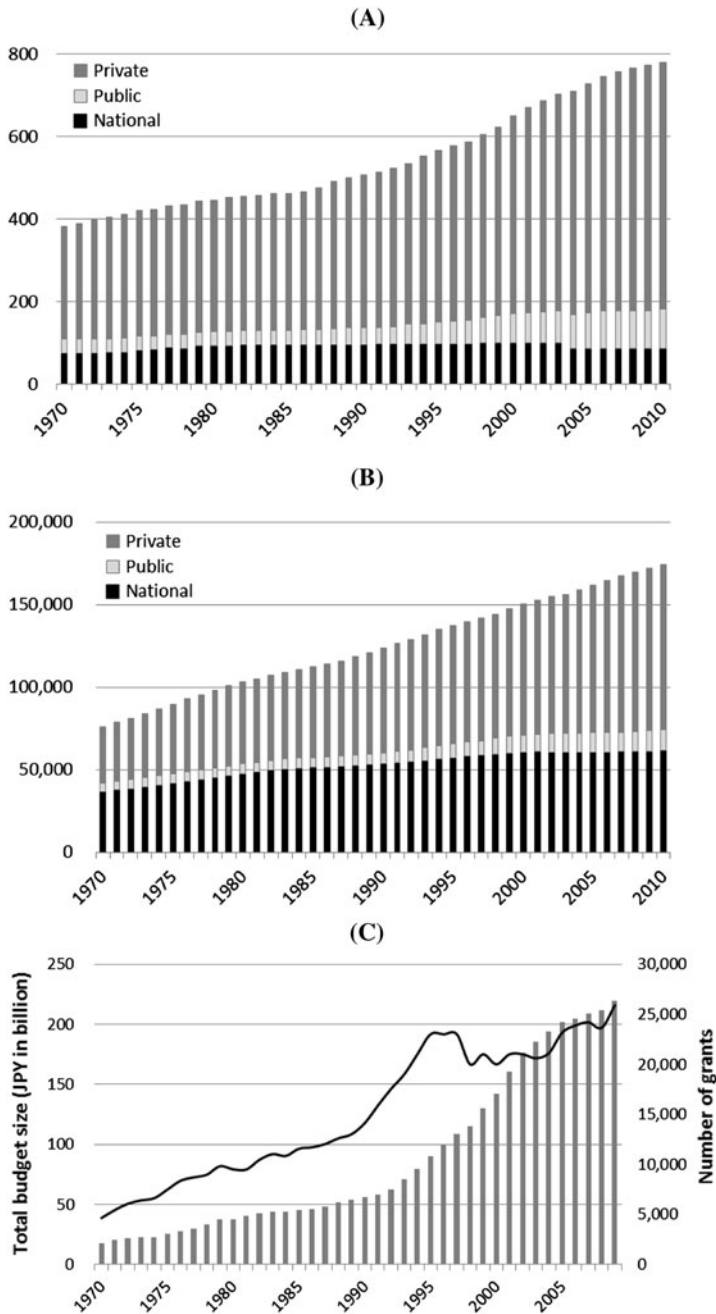


Fig. 1 Growth of university system: **a** number of universities, **b** number of faculty members, and **c** Grants-in-Aid for Scientific Research. *Source:* **a** and **b**. School Basic Survey conducted by the Ministry of Education, Culture, Sports, Science and Technology of Japan (MEXT). **c**. The white papers of MEXT (http://www.mext.go.jp/b_menu/hakusho/html/kagaku.htm) and the list of GIA grant projects edited by Gyosei. The price level is adjusted by the GDP deflator (the base year is 2000)

example, national universities account for 71% of the number of Ph.D. graduates as of 2009,¹ and the main funding system called Grants-in-Aid for Scientific Research (its detail is explained later) provides 67% of its total budget for national universities and 33% for other universities and institutions as of 2010.² Among the national universities, the top seven universities (Universities of Tokyo, Kyoto, Tohoku, Osaka, Hokkaido, Kyushu, and Nagoya) have enjoyed especially prestigious status, taking the initiative both in research and in education since the late nineteenth century. Because of their importance and data availability, this section and a part of the following analyses focus on national universities or the top seven universities.³

In terms of budgetary structure, the national universities, on average as of 2008, obtained their revenue as block grants from the government (40%), operating income (tuition fees, income from affiliated hospitals if any, etc.) (40%), and competitive funds (20%) (Center for National University Finance and Management 2010: Chap. 14). The first and the second parts are spent mainly on salary for faculty members, education, and other operating costs, while the competitive funds play the major role in research activities. In most countries, the revenue of universities consists of organization-level and individual- (or project-) level funds, which is often called a dual-support system (Geuna and Martin 2003). The organization-level funds are often block grants from governments and are spent on ordinary expenses, while the individual-level funds tend to be used for specific purposes such as research projects. The Japanese system is no exception in this respect.

It is not straightforward to draw a precise picture of the funding structure specifically for research activities because the relation between revenue and expenditure is often unclear. However, a prior study based on an individual-level survey offers an overview. Center for National University Finance and Management (2009: Chap. 6) shows that individual researchers in national universities obtained their research budget primarily from the governmental competitive funds at the individual (or project) level (65% as of 2007), while a minor part of the budget is supported by their university (20% as of 2007) and industrial funds (5% as of 2007). The budget from universities is basically distributed equally with limited consideration of individual performance. The proportion of each funding source can vary with scientific fields; for example, the university budget is more important in social sciences than in natural science. Nevertheless, governmental funds for individual researchers have played the principal role in university research. This emphasis on individual-level funds has been strengthened since the 1960s (Center for National University Finance and Management 2009: Chap. 6).

Currently, there are numerous types of governmental research funding systems, but among others, Grants-in-Aid for Scientific Research (GIA, hereafter) have been playing the most fundamental role. GIA is the largest funding system, covering all scientific fields from social sciences to natural sciences, and is broadly awarded to university researchers from Ph.D. students to full professors.⁴ The system of GIA dates back to the 1950s. Since then, the budget size of GIA has been consistently increased in an attempt to improve the capability of university research, and it amounted to 200 billion JPY (2.2 billion USD) in 2010 (Fig. 1c). Until the 1980s, GIA had been virtually the sole individual-level funds

¹ Source: School Basic Survey conducted by MEXT.

² Source: Japan Society for the Promotion of Science (<http://www.jsps.go.jp/j-grantsinaid/index.html>).

³ The budgetary and funding structure of private universities is somewhat different from that of national universities. Kneller (2010) and Asonuma (2002) also give an overview of the revenue and expenditure structure of Japanese universities.

⁴ The general information of GIA is given in the MEXT website (http://www.mext.go.jp/a_menu/shinkou/hojyo/main5_a5.htm).

(Center for National University Finance and Management 2009: Chap. 6). Afterwards, the government has implemented many funding systems for various objectives.⁵ Nevertheless, GIA accounts for the majority of all the competitive funds (National Institute of Science and Technology Policy 2009), and its role as the primary research funds has unchanged. For this central role in academic research and the availability of data, this study focuses on the distribution of GIA.

Data source

The grant information for GIA is available through a public database.⁶ This database provides the information of research projects supported by the system of GIA since 1965; such as grant amount, project members, a grant category, a scientific field, and resulting publications. I obtained the whole dataset as of 3 Dec 2008 and reconstructed it for this study. The final dataset includes approximately 600,000 grants and 210,000 university researchers.

In the analysis of funding inequality at the individual level, the total number of researchers has to be known. In this respect, an obvious limitation of the database is that it does not include non-grantees. It is still plausible to argue that non-grantees are not researchers (In fact, faculty members receiving no GIA grant tend to be teaching staff), but this can overlook some actual researchers (e.g., those who need very small amount of research funds). National statistics provide the number of university faculty members (Fig. 1b), but they do not distinguish faculty members who actually do research and those who mainly engage in non-research missions. Thus, simply taking the number of all faculty members would overestimate the inequality due to those who do not research. This problem is less serious for national universities than for private universities, and is even limited for top universities. The majority of faculty members in top universities actually engage in research activities and obtain the GIA grants.⁷ For these characteristics of universities, I focus on national universities or the top seven universities in some of the following analyses.

Gini coefficient

In order to quantify the funding distribution, I use the Gini coefficient as an indicator of inequality. It is a simple measure ranging from zero (when everyone takes the same amount) to one (when one takes all). The Gini coefficient has been broadly used in scientometric research (e.g., Burrell 1991; Zitt et al. 1999). It has several advantageous features such as scale independence (the total amount of funds does not matter) and population independence (the number of universities or researchers does not matter) (Ray 1998). Mathematically, the Gini coefficient is defined as the ratio of *the area between the line of equality and the Lorenz curve* divided by *the total area under the line of equality*, where the Lorenz curve shows the percentage of the total wealth given to the bottom x% of

⁵ As of 2008, there are 44 competitive funding systems. Some of them are individual-level funds for university research, while others are awarded to institutions, primarily focus on industrial research, or aim at education.

⁶ National institute of informatics (<http://kaken.nii.ac.jp/en/>).

⁷ I examined the ratio of grantees in all full and associate professors in the University of Tokyo (ranked first). After excluding those who have no publication, part-time employees (many of them are from industry), and apparent non-researchers, 95% or more of professors have obtained the GIA grants regardless of scientific fields.

entities (Dorfman 1979) (see Fig. 2b). To calculate Gini coefficients for the funding distribution, I use the following equation (Halfman and Leydesdorff 2010):

$$\text{Gini coefficient} = \frac{\sum_{i=1}^n (2i - n - 1)y_i}{n \sum_{i=1}^n y_i},$$

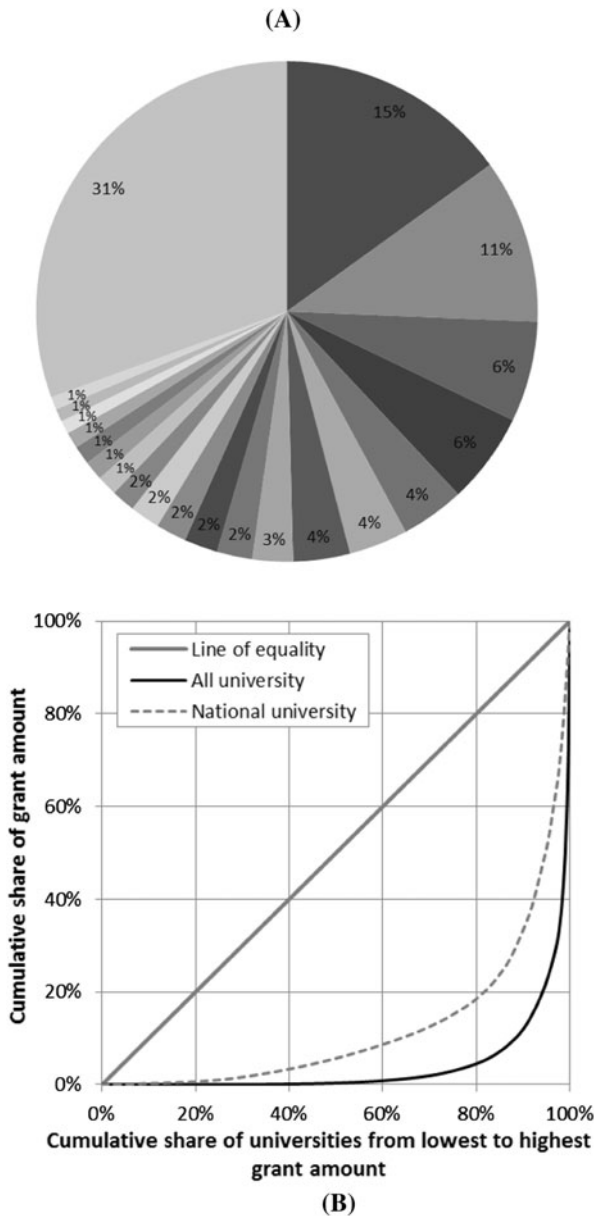


Fig. 2 Grant share at the university level: **a** share of the top 20 universities and the rest in 2005 and **b** Lorenz curve for the grant share of universities in 2005

where n denotes the total number of researchers or universities and y_i ($i = 1$ to n ; $y_i \leq y_{i+1}$) denotes the amount of the grant awarded to the i th researcher or university.

The Gini coefficient can take a broad range of values depending on contexts. Since there is no absolute standard, a value of the coefficient is difficult to evaluate of itself but should be considered with its context. This is why the current study attempts to present the results in comparative manners. Just for reference, the income disparity in developed nations in the late twentieth century ranges from 0.2 to 0.5, among which the U.S., known as highly unequal society, took 0.46 of Gini coefficient in 2008 (Denavas-Walt et al. 2009). Thus, Gini coefficients greater than this range seem to imply high inequality.

University level distribution

First, I examine the university-level funding distribution. Figure 2a illustrates the share of the top 20 and the rest of the Japanese universities in 2005. I calculated the grant amount in each university by summing up the grants awarded to principal investigators (PIs)⁸ affiliated with the university. Among all 726 universities, 218 universities received no grants, most of which are private schools. The pie chart indicates that the top university took 15% of the total amount and that the top seven universities accounted for 50%. The top 20 universities took nearly 70%, while the rest of 30% was distributed to the rest of hundreds of universities.

Figure 2b shows two Lorenz curves of the university-level funding distribution in 2005; one for all universities and the other for only national universities. It clearly shows high inequality with a Gini coefficient of 0.919 for all universities. I also examined the transition of the Gini coefficients and found that the extent of inequality has not noticeably changed over 30 years (0.900 in 1995, 0.899 in 1985, and 0.914 in 1975). As mentioned above, many private universities are not research oriented, and the number of private universities has significantly increased in the periods (Fig. 1a). This might have raised the Gini coefficients in recent years, so I also calculated Gini coefficients only for national universities, which have constantly received approximately 80% of the total budget. The Gini coefficients of each year are 0.760 in 2005, 0.787 in 1995, 0.788 in 1985, and 0.754 in 1975, which also indicates limited change during the four decades.

Individual level distribution

Multiple awarding and large grants

Next, I examine the individual-level funding distribution. Japanese policy makers have been discussing the individual-level inequality in recent years. Above all, two issues have been of particular interest. The first is that the size of some grants may be excessively large. Since the 1990s, the government has enriched the investment for so-called big science, contending that big science enables Japan to take the initiative in globalized science (MEXT Subcommittee of Academic Science 2003). Many individual scientists perceived this trend as increasing inequality although the government denied this criticism.

⁸ A PI is the researcher who supervises a certain project. In most projects, there is only one PI, but for large projects, there may be more than one PI. In such cases, grant amount is divided by the number of PIs to calculate the per-PI amount.

The second issue is that some researchers obtain too many grants simultaneously. The government recognized the multiple awarding as a problem to be solved, and has set several rules that prohibit a researcher from having multiple grants for similar purposes (MEXT Subcommittee of Academic Science 2008).

For the former issue, I examine the transition of grant size since the 1970s. Figure 3a distinguishes three size groups and shows their proportions: large (greater than 100 million JPY), middle (10–100 million JPY), and small (less than 10 million JPY). The graph clearly indicates the tendency to enrich larger grants. In 1976–1980, small-size grants account for 66% and middle-size grants account for 34%, whereas small size is only 32% but middle size is 46% and large size is 21% in 2001–2005. Noticeably, since the 1990s, large-size grants have constituted about 20% of the total budget. However, the recent two periods show a slight decrease in the proportion of large grants (from 23% in 1996–2000 to 21% in 2001–2005), while the budget amount for large-size grants have increased by 35%. This may reflect the revision of the governmental funding policy to address the criticisms on excessively large grants.⁹

As for the second issue, one researcher can apply for and obtain more than one grant simultaneously although there are some restrictions. This issue has aroused criticism also in other nations (e.g., the U.S.: Hand 2008). A typical argument against multiple awarding is that researchers can devote only limited efforts to each granted project, which compromises the efficiency of the fund use. In addition, when multiple awarding is allowed, researchers are incentivized to apply for many grants, which inevitably increases the time for grant application and decreases the time for actual research. To scrutinize this issue, I counted the number of grants that each researcher received as a PI simultaneously in a single year¹⁰ (Fig. 3b, c). I considered only full and associate professors in this analysis because other junior researchers are unlikely to receive multiple grants.¹¹ Figure 3b, taking grantees from all universities as a population, shows that most grantees obtained only one grant and 10–20% received two or more grants simultaneously. The ratio of multiple awarding increased from 9.5% in 1975 to 22% in 1995. During this period, the number of grantees had tripled (Fig. 1c). Thus, while the GIA grant had become more common, multiple awarding had also prevailed. Since then, the ratio of multiple awarding has slightly decreased to 16% in 2005.¹² This may be the result of the government's effort to address the criticism on multiple awarding, but the number of such incidents has actually slightly increased. In addition, when both PIs and non-PI members are considered, 42% researchers were involved in more than one project in 2005. The maximum number of multiple awarding to a single PI was seven and that to a single PI or member was 27 in 2005. In order to take non-grantees into consideration, Fig. 3c takes the population of all faculty members (full and associate professors) in the top seven universities whether they obtained grant in a certain year or not. It clearly indicates that the GIA funds have been

⁹ In fact, the government mentions that the average grant size has been decreasing since 2002 (http://www.mext.go.jp/a_menu/shinkou/hojyo/1289168.htm).

¹⁰ For example, if a researcher received one grant from 1993 to 1995 and another grant from 1995 to 1997, this researcher is regarded as a multiple awardee in 1995.

¹¹ In Japan, typical laboratories consist of one full professor, one associate professor, a few junior staff (assistant professors, postdoctoral researchers, etc.), and students, where full and associate professors control the management of the laboratory and often secure research funds for other staff.

¹² In 2005, approximately 4,000 full and associate professors received two or more grants as a PI, which accounts for 16% of all the professors who received at least one grant in the year. I ignored non-grantees in this analysis because the true number of all researchers is not known. If all professors in all universities are considered (which include non-researchers), the ratio drops to 5.4%.

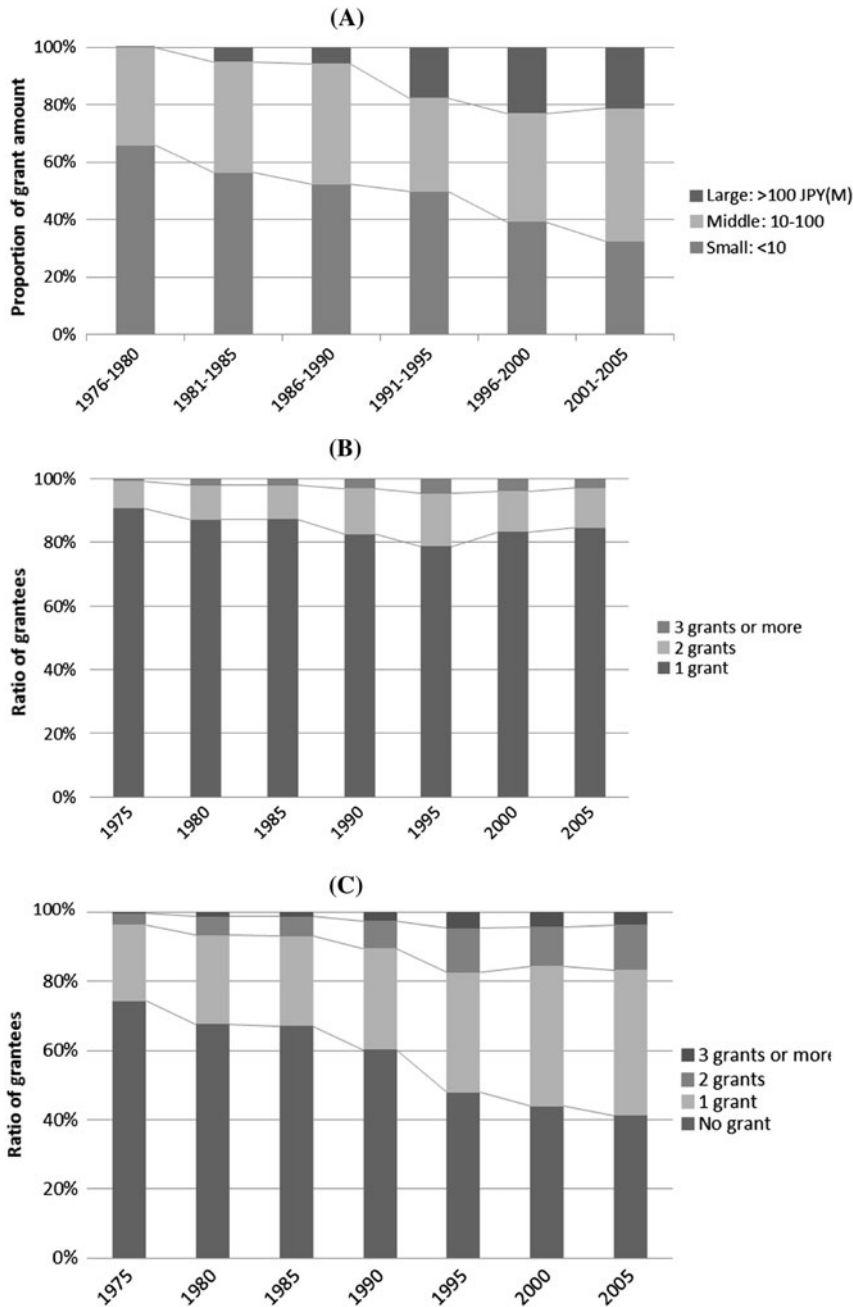


Fig. 3 Grant size and multiple awarding: **a** grant size, **b** multiple awarding (all universities), and **c** multiple awarding (the top seven universities). **a** The grant amount during 5 years are summed up for each size group. The price level is adjusted by the GDP deflator (the base year is 2000). **b** and **c** Only full and associate professors are considered. Non-grantees are not included in **b** while they are in **c**

distributed increasingly broadly (i.e., the ratio of no grant has largely decreased) whereas multiple awarding had also become more prevalent until 1995. As of 2005, 17% of full or associate professors in the top seven universities received two or more grants simultaneously.

Inequality of distribution

To examine the overall distribution, I calculate the Gini coefficient at the individual level. As mentioned above, a problem in this analysis is that the true number of “researchers” is not known. To mitigate this limitation, I draw on the following two populations. The first analysis takes all university researchers but only grantees as a population, assuming that all “researchers” receive the GIA grant at least once in a sufficiently long period.¹³ Put differently, faculty members who do not receive any grant over a long period of time are regarded as a non-researcher. This assumption may be too simplistic but can be still reasonable in that GIA is the primary funds for university research and that most faculty members in research-intensive universities are actually awarded the grant. Nevertheless, Gini coefficients obtained in this way can be underestimation due to the potential exclusion of actual researchers who happened not to receive any grant in a specific term. The second population focuses on the top seven universities. The seven universities have constantly obtained approximately 50% of the GIA budget. Faculty members in the seven universities accounted for 30% of those in all national universities and 11% of those in all universities as of 2005. Based on the assumption that most faculty members in these universities are researchers, I use the number of faculty members reported by each university as a population. This calculation has an advantage of including researchers who happened to receive no grant in a certain term, but it can include some non-researchers, resulting in the overestimation of inequality.

Figure 4a describes Lorenz curves for the two above-mentioned populations. In this analysis, I also focus on full and associate professors because other junior researchers are institutionally awarded only limited amount of grants. The Gini coefficients are 0.685 (all universities) and 0.746 (top seven universities) in 2001–2005.¹⁴ Figure 4b illustrates the transition of the inequality since the 1970s. The two lines are more or less stable around 0.70 and 0.75 respectively over the four decades, but the inequality seems to have increased in the 1970s and the 1980s and then decreased in the 1990s and the 2000s. The slight decrease in the recent decade might be attributable to the governmental policy to suppress inequality.

Since grant size greatly differs by scientific fields, I also examine the funding distribution by field. On the basis of the research areas of faculty members, I divided all researchers into five groups.¹⁵ I additionally illustrate the transition of the size of the five fields in Appendix. Figure 4c reveals the difference of inequality between fields. The lowest inequality is found in social sciences (Gini coefficient = 0.595). Compared to this,

¹³ I use 5 years in the following analysis. The government regards 5 years as a standard term of research. The acceptance rate of GIA has been approximately 25%, so most applicants are supposed to receive at least one grant in 5 years if multiple awarding is sufficiently limited.

¹⁴ For the top seven universities, faculty members who received no grant are taken into consideration. They account for 16% of all faculty members in 2001–2005. On the other hand, for all universities non-grantees are ignored. Note that this different definition of populations results in larger inequality in the top seven universities than in all universities.

¹⁵ The field of each researcher is determined on the basis of the field most frequently assigned to their past grants.

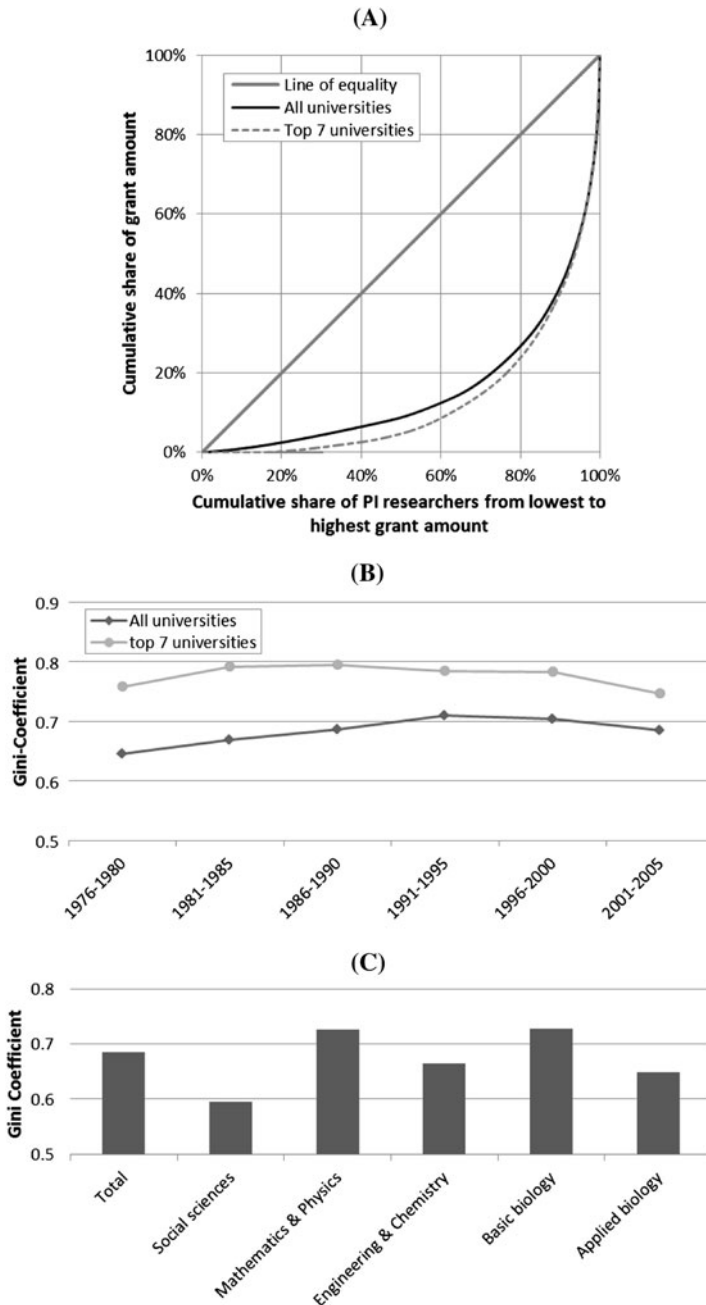


Fig. 4 Grant share at the individual level: **a** Lorenz curve of the grant share of individual researchers in 2001–2005, **b** transition of inequality, and **c** Gini coefficients by field in 2001–2005. For the two populations, all the grants awarded to each individual in 5 years are summed up. Each grant amount is divided by the number of PIs if more than one PI is assigned to one grant. Only full and associate professors are considered. In **c**, the population of only grantees from all universities is used

the four natural science fields show higher inequality. This is possibly because the size of grants in social sciences tends to be smaller. Among natural sciences, basic biology is the most unequal (Gini coefficient = 0.728). Mathematics and physics shows the second highest inequality (0.726), and the other two fields have relatively low inequality (engineering and chemistry: 0.664 and applied biology: 0.649). Thus, the grant distribution in basic research seems more unequal than that in applied research.

Transition of inequality across cohorts and over careers

Next, I investigate the transition of inequality over researcher's career with controlling the cohort. The limitation of the analysis in the previous sections is that the population of researchers includes both less experienced researchers and more experienced researchers, whose fund sizes may well be different. Thus, I examine inequality focusing on four cohorts of researchers who started their career in (1) 1966–1970, (2) 1971–1975, (3) 1976–1980, and (4) 1981–1985. Then, I follow the transition of inequality in each cohort over 25 years.¹⁶ The left column of Fig. 5 shows the transition of average grant sizes for cohorts 1 and 4, while the right column illustrates the transition of Gini coefficients for the four cohorts.

The left column indicates a growing fund size for cohorts 1 and 4 in five fields. The pattern of growth differs from field to field. In terms of the growth throughout a career for the same cohort (see the symbols on the line graphs), social sciences experience a significant growth only in the early career stage, and mathematics and physics shows similar pattern. In other fields of natural sciences, the growth continues mostly throughout the career. Comparing the two cohorts (see the symbols above the line graphs), applied biology and engineering and chemistry show that the average fund size for cohort 4 is larger than that for cohort 1 in all career stages. This seems to suggest that average spending in these fields have simply increased between two cohorts. On the other hand, social sciences show significant differences only in the first and second career stages, which implies that young researchers have come to be awarded larger funds but that the fund size for experienced researchers have not changed substantially. Basic biology and mathematics and physics show significant differences at the first and last career stages, which indicates a growing investment for young researchers and the most experienced ones.

The left column also shows the deviation of fund size by error bars (25 and 75 percentiles). The deviation increases quickly as a career advances and is larger in cohort 4 than in cohort 1. These may appear to imply increasing inequality over the career and over the generations, but this is not obvious because the total funding budget has grown simultaneously (Fig. 1c). With this regard, the Gini coefficient is a useful indicator of inequality because it is scale independent. The right column of Fig. 5 indicates increasing inequality over the career, but it does not indicate that the level of inequality increases over generations. The increasing inequality over career generally suggests that researchers are given relatively equal opportunity in their junior stage, and that winners in the stage take more resources in the subsequent stages. The inequality at the first career stage varies largely with fields. While social sciences, engineering and chemistry, and applied biology show relatively low inequality (around 0.4), mathematics and physics indicate high inequality (around 0.6) from the beginning. Interestingly, basic biology had a high

¹⁶ I chose the researchers who have 25 or more years of career. That is, I excluded researchers whose career ends before 25 years; such as those moved to industry or education-oriented universities. If these researchers were included, the inequality in later career stages would be higher.

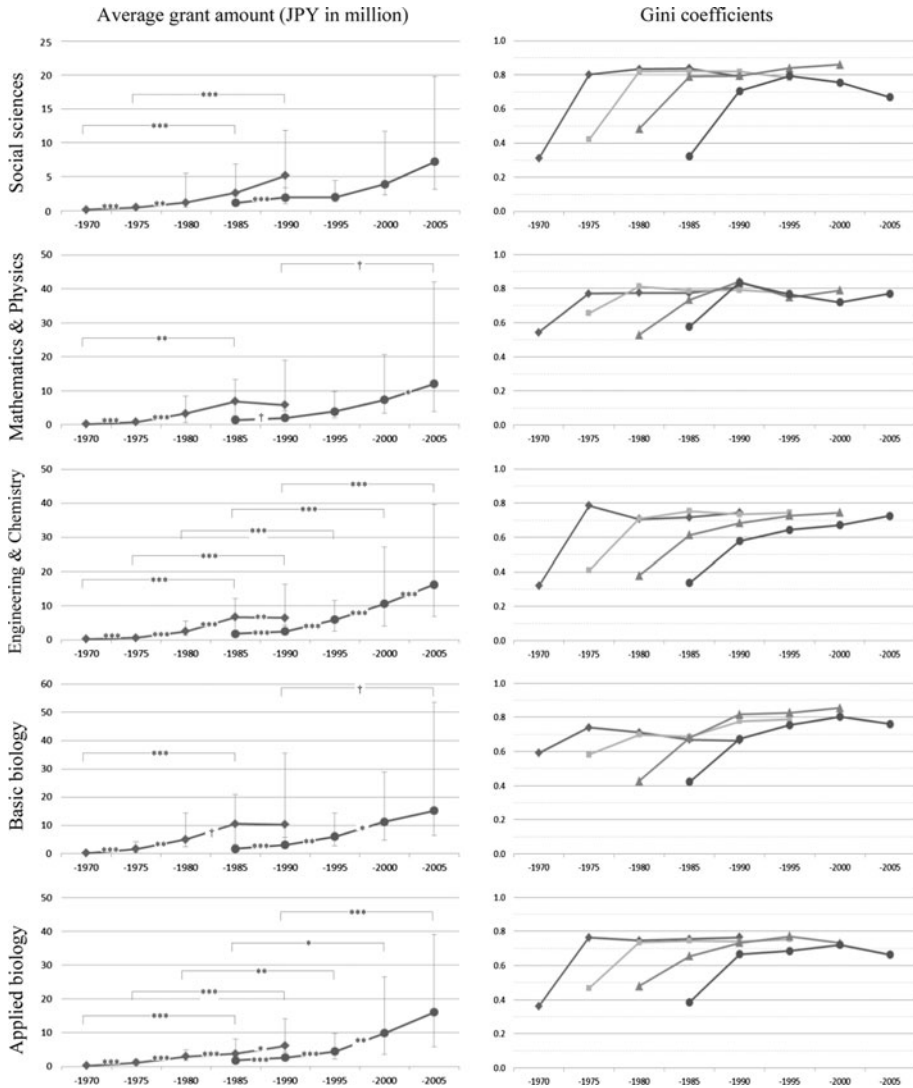


Fig. 5 Transition of grant distribution across cohorts and over career. I chose the researchers who have 25 or more years of career (the latest grant-awarded year minus the earliest year is 25 or more) and whose first grant was given when they were an assistant or post-doctoral researcher. The left column indicates the median grant amount (total in 5 years) for two cohorts whose career started during 1966–1970 and 1981–1985. Error bars indicate 25 and 75 percentile amounts. The scales of vertical axes differ between fields. Significant differences are indicated by symbols (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.1$) for the mean grant size between two consecutive career stages in the same cohort (*on the line*) and that between the two cohorts in the same career stage (*over the line*). The right column shows the transition of Gini coefficients over the career in the four cohorts whose career started during 1966–1970, 1971–1975, 1976–1980, and 1981–1985, respectively

inequality (around 0.6) at the first career stage in cohorts 1 and 2, but this largely decreased in cohorts 3 and 4 (around 0.4). This may suggest that the field of basic biology had come to place a greater emphasis on equal opportunity at the junior stage. At the last career stage, on the other hand, the level of inequality is less diverse (0.7–0.8) across fields. As for the

transition over the career, cohorts 1 and 2 show a quick growth of inequality after the first career stage, so the performance at the very beginning used to be critical for the later career. However, cohorts 3 and 4 show relatively gentle slopes, which might imply that longer time is allowed in the process of identifying excellent researchers.

Inequality in output

To consider the efficiency of funding distribution, I also examine the distribution of output in terms of the number of publications. The grant database archives the reports of each granted project, which list the publications from the project. First, I counted the number of publications in each university, whereby the Gini coefficient is calculated. In 2005, the coefficient for all universities is 0.845, which is slightly smaller than that for funding distribution, or 0.919. I also focus on national universities and compare the inequality. The Gini coefficient for the publication distribution is 0.676, while that for funding is 0.760.¹⁷

Furthermore, I compare the individual-level distributions of funding and publications. The Gini coefficient for the publications of full and associate professors (excluding non-grantees) in all universities in 2001–2005 is 0.592, which is smaller than that for funding, or 0.685 (Fig. 6a). I also draw on the population of all full and associate professors (including non-grantees) in the top seven universities, which results in the Gini coefficient of 0.645 for publication and 0.746 for funding. Finally, I compare the two distributions in each scientific field. Figure 6b indicates that publication inequality is smaller than funding inequality in all fields except for social sciences. The difference is especially noteworthy in mathematics and physics (0.523 vs. 0.726) and basic biology (0.570 vs. 0.728). Overall, the greater inequality in funding than in publication seems to imply an excessive concentration of funding.

Discussion

Using a Japanese grant database, this study examines the distribution of research funds from several perspectives. First, the data indicates high inequality at the university level (0.919 for all universities and 0.760 for national universities) and that the extent of inequality has not been largely changed during 1975–2005. However, the number of faculty members is already highly unequal in Japanese universities (e.g., the Gini coefficient in 2005 was 0.602), so the high inequality of funding itself needs to be carefully interpreted. This study also examines the distribution of publications and indicates that the publication distribution is less unequal than the funding distribution. This seems to imply an overconcentration of funding in limited universities.

Next, this study examines the individual-level inequality. The data shows that individual-level funding inequality is slightly moderate compared to the university level. During the last 30 years, the funding inequality has been more or less stable. The slightly decreasing inequality in recent years may be attributable to the government's recent effort to suppress excessive inequality. To disentangle the causes of inequality, this study analyzes the transition of funding distribution from several perspectives. In the past 30 years, the total budget for the GIA funds has increased by 20 times (after controlling the price

¹⁷ For validation, I counted the number of publications whose affiliation includes each national university using Web of Science, which resulted in a similar level of Gini coefficient.

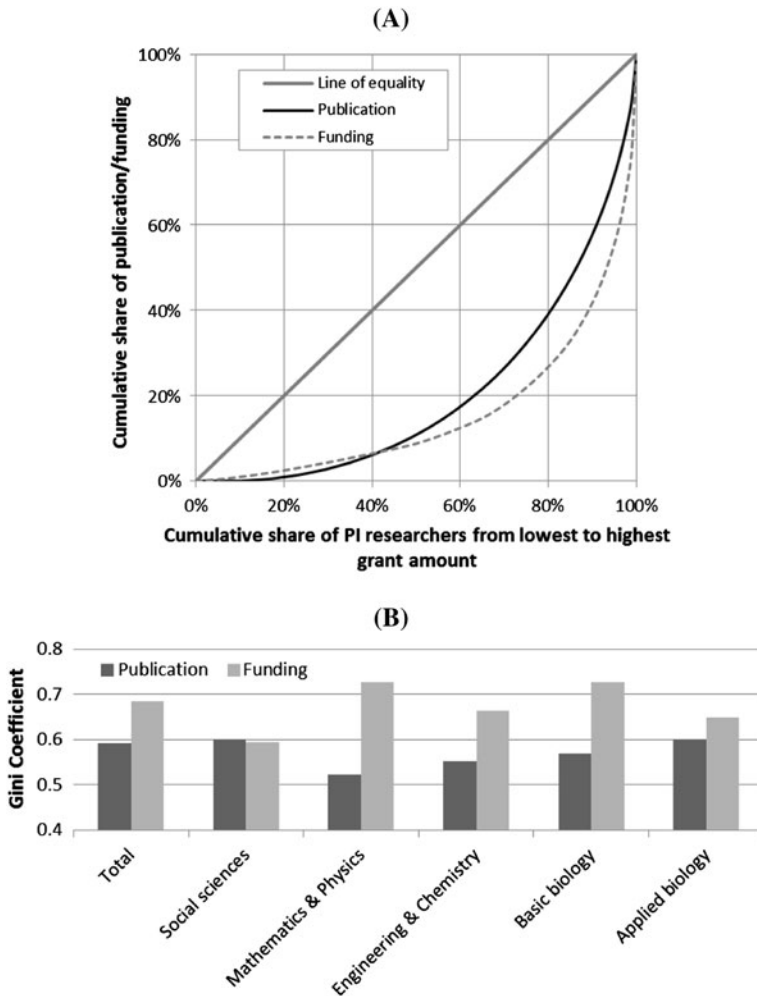


Fig. 6 Comparison of inequality between output and input: **a** Lorenz curve for the publication and grant share of individual researchers and **b** Gini coefficients by field. For full and associate professors in all universities, the distributions of funding and publications in 2001–2005 are illustrated. I counted the number of publications resulting from the grants, and summed them up for each PI. When more than one PI is assigned to one grant, the number of publications is divided by the number of PIs (but this does not change the Gini coefficients substantially)

level) and the number of grants has increased by 6 times. Simultaneously, the size of academia has increased by 2–3 times in terms of the number of universities or faculty members. One clear feature of funding policy in this period was an inclination to favor larger grants. Especially, after the 1990s, the large-size grants (10 million JPY or greater) have accounted for about 20% of the total budget. Since large grants have been criticized for presumable inefficiency, the relation between grant size and the productivity of grantees should be further investigated. In addition, this study examines multiple awarding. The data shows that 16% of grantees played the PI role in two or more projects in 2005. Specifically in the top seven universities, 17% of faculty members received multiple grants

while 41% received none in 2005. Considering diverse objectives of different grant categories, multiple awarding may remain to some extent. Nevertheless, it can be inefficient for a researcher to simultaneously receive many grants and put limited efforts into each granted project. This possibility should be thoroughly studied in future research. Next, the difference of funding inequality across scientific fields is examined. The data suggests that natural sciences have greater inequality than social sciences and that basic research has greater inequality than applied research. The transition of inequality over career stages is also examined with cohorts being controlled. As expected, the data shows higher inequality in senior stages than in junior stages, implying that the success at the earlier career determines future research input. This tendency holds in all fields and in all cohorts. A few interesting features in a specific field are indicated. For example, in social sciences, the inequality at the beginning of a career is fairly low but rapidly rises to its maximum. In mathematics and physics, the inequality is relatively high from the beginning of a career. In basic biology, the inequality at the junior stage decreased from the 1970s to the 1980s, seemingly favoring more equal chance for young scholars. Finally, the funding distribution is compared with the publication distribution. The results indicate greater inequality in funding than in publications in all fields except for social sciences. Although some literature has suggested that funding distribution may well be unequal in accordance with the inequality of output (Hicks and Katz 2009), this study seems to indicate a potential overconcentration of funding.

This study explores the inequality of research funds from several perspectives, but it entails some limitations that future research should address. First, this study depends only on GIA, one system of research funds. GIA is absolutely the largest and most fundamental funding system in Japan, but university researchers can raise funds from other governmental agencies, industry, and their university. Future research should consider these other sources of budget. Second, “researchers” and “research universities” are not easy to determine clearly, so Gini coefficients can vary depending on the selected population of researchers or universities. At the university level, many national universities are research-intensive and many private universities are education-oriented, whereas there are universities where some professors engage in research and others do not. Thus, it is controversial which universities should be included in the analyses. At the individual level as well, it is not easy to identify faculty members who actually do research. The first assumption I used in this study, that researchers are supposed to obtain at least one grant from GIA in a certain term, may not be far from the truth, but it can exclude some actual researchers. The second assumption, that all faculty members in the top seven universities are researchers, may be nearly the case. However, the cost of ignoring other many universities is not negligible, and we still have to be careful that there are non-researchers even in the top universities. Third, for the argument of efficient funding, the final section of this article considering both input and output is of greater interest, but the analysis in this study is only preliminary. Future research should look into the causality between funding and scientific production. In fact, the effect of funding has been examined by only a limited number of studies (e.g., Crespi and Geuna 2008). I expect that the GIA database developed in the current study can contribute to this line of research. Furthermore, with the database, the efficacy of excessively large grants and multiple awarding, for example, can be examined in a more rigorous manner.

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Appendix

See Fig. 7.

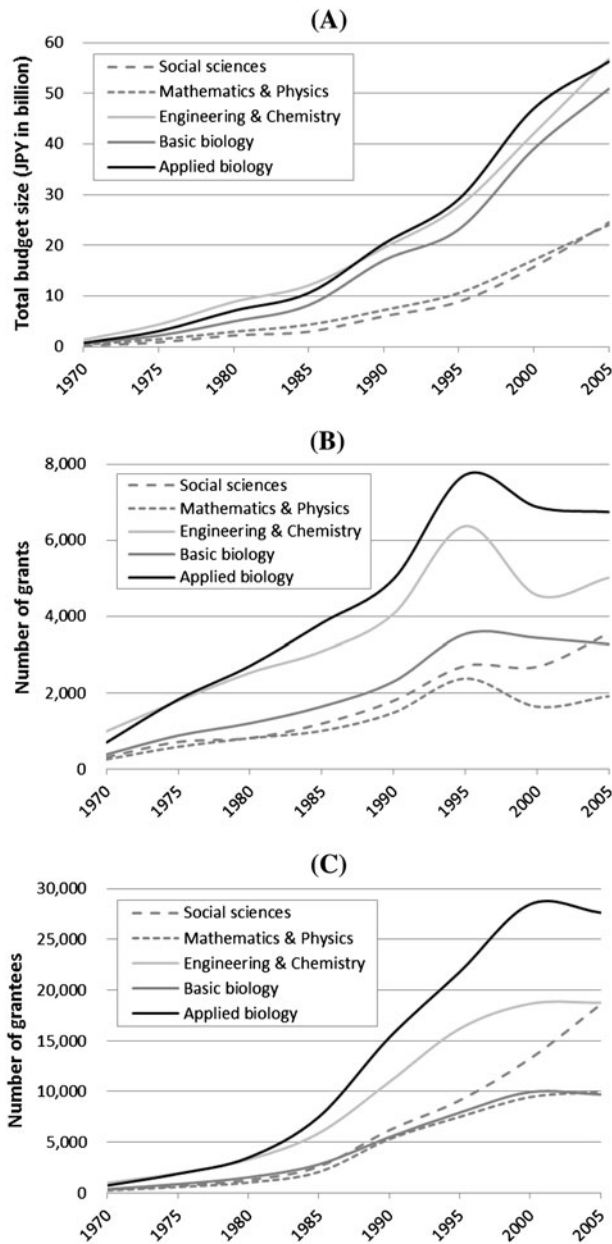


Fig. 7 Growth of university system by field: **a** total budget, **b** number of grants, and **c** number of grantees. **a** The amount of grants awarded to researchers is summed up for each field. The price level is adjusted by the GDP deflator (the base year is 2000). **c** Since the statistics for university researchers do not distinguish scientific fields, I present the number of grantees (instead of all researchers) in each field

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